MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 4D21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Vertical, base down or up

Weight: 6.5 ounces (184 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec1	Ec2	Ib	Pg1	Pg2	Pp	Cooling	T(seal)	TE
Unit:	MHz	V ac	V dc	V dc	V dc	mA dc	W	W	W	---	°C	°C
Maximum:	110	5.25	3,000	-500	400	225	5	20	125	---	200	225
Minimum:	---	4.75	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	5.0	2,500	Adj	500	50	---	---	---	Note 1	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	2					
1301	Filament current	-		If	6.4	7.2	A ac
1231	Peak emission	-	eb = ec2 = ec1 =2,500 v	is	3.5	---	a
1261	Electrode voltage (grid)	-		Ec1	-60	-83	V dc
1266	Total grid current	3		Ic1	---	-10	µA dc
1266	Primary grid emission (control)	-	Ef = 5.5 V ac; t = 15 seconds; Pg1 = 6 watts or Ic1 = 85 mA dc; anode and screen grid floating	Isg1	---	-200	µA dc
1266	Primary grid emission (screen)	-	Ef = 5.5 V ac; t = 15 seconds; Ec1 = 0; Pg2 = 25 watts or Ic2 = 75 mA dc; anode floating	Isy2	---	-200	µA dc
Ⓓ 1372	Current division (method 8)	-	Eb = 500 V dc; Ec2 = 350 V dc; Ec1 = -150 V dc; egk/ib = 800 ma; tp = 2 µs (min); prr = 30 (min)	egk ic1 ic2	--- --- ---	150 160 300	v ma ma
	Quality conformance inspection, part 2						
1316	Amplification factor	-	Connect g1 to g2; Ec2 = 300 V dc; Ic2 = 60 mA dc; anode floating	Mu	5.2	6.5	---
1236	Power oscillation (1)	4	Eb = 3,000 V dc; Ec2 = 350 V dc; Ib = 100 mA dc; Rg1 = 15,000 ohms; F = 15 MHz	Po Ic2	175 15	--- 50	W (useful) mA dc
1331	Direct-interelectrode capacitance	-		Cgp Cin Cout	--- 9.2 2.5	0.07 12.4 3.5	pF pF pF
1031	Low-frequency vibration	-	No voltages	---	---	---	---
1042	Shock, specified pulse	-	No voltages; test condition A, except 15G	---	---	---	---
---	Low-frequency vibration and shock, specified pulse-test end points:						
1261	Electrode voltage (grid)	-		Ec1	-60	-83	V dc
1266	Total grid current	-		Ic1	---	-10	µA dc

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
---	Life test	4	Group C; Eb = 3,000 V dc; Ec2 = 350 V dc; Ib = 100 mA dc; Rg1 = 15,000 ohms; F = 15 MHz (min)	t	500	---	hrs
---	Life-test end points:						
1231	Peak emission	-		is	3.0	---	a
1266	Primary grid emission (control)	-		Isg1	---	-200	µA dc
1266	Primary grid emission (screen)	-		Isg2	---	-200	µA dc
1266	Total grid current	-		Ic1	---	-10	µA dc
1236	Power oscillation (2)	4,5	Eb = 3,000 V dc; Ec2 = 350 V dc; Ib = 100 mA dc; Rg1 = 15,000 ohms; F = 110 MHz	Po	175	---	W (useful)

NOTES:

1. During all electrical tests involving application of filament power, forced-air cooling of the TUT is allowable and a heat-dissipating connector (EIMAC HR-6, or equivalent) may be used on the anode terminal.

2. The AQL for each test listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

3. This test shall be the first test performed at the conclusion of the holding period.

4. The value of Rg1 = 15,000 ohms shall apply when a self-excited oscillator circuit is used. If an amplifier circuit is used, fixed bias or a combination of fixed bias and grid-leak resistance may be used.

5. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

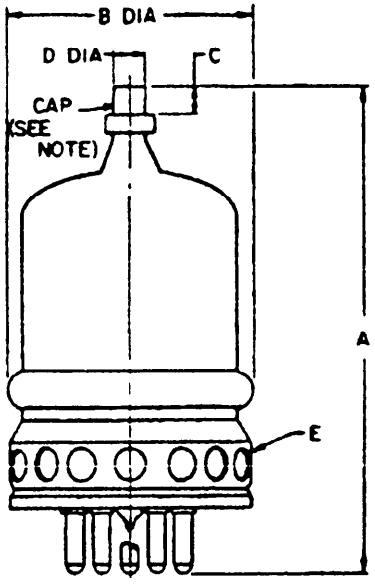
User activities:
Army - AR, ME
Navy - AS, OS, MC, CG
Air Force - 11

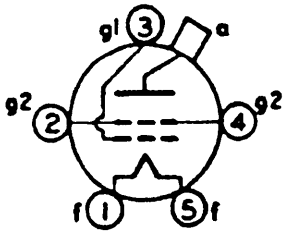
Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5960-3287)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.188 (131.78)	5.688 (144.48)
B		2.812 (71.42)
Quality conformance inspection, part 3 (See note 5)		
C	.328 (8.33)	
D	.350 (8.89)	.365 (9.27)
E	A5-97 (EIA)	

NOTE:
Does not include increase due to solder.

FIGURE 1. Outline drawing of electron tube type 4D21.

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